

Tone Relative to Substance in European ESG Disclosure: A Continuous Corpus-Relative Score, 2018–2024

1 Scope and contribution

This paper measures how the tone of a report’s ESG disclosure stands relative to the substance of that disclosure. We compute the measure for 343 reports of 49 listed firms from seven euro-area countries, one report per firm and year over 2018–2024 (§2). Tone is the polarity of the extracted ESG text under the finance dictionary of Loughran and McDonald (2011); substance is the density, in the same text, of mentions of a documented ESG vocabulary (§3).

The measure is a continuous score, and it is relative to the corpus. Each component is standardised over the 343 reports, so a value locates a report within this corpus and has no meaning outside it; its sign says only whether the report lies above or below the corpus mean (§3.4). The idea behind the score, positive language in excess of the substance that supports it, fixes which direction of the score is of interest.

Two scores. The main score, ESGSI (1), follows the architecture of Lagasio (2024), a tone component minus a substance component, each rescaled across the sample; we keep that architecture and correct the specification of its parts, as listed below. The second score, the Extended ESGSI $\text{ESGSI}^{\text{ext}}$ (2), is the new metric this paper proposes. It adds quantification on the side of substance and hedging on the side of tone, with weights that are a choice rather than an estimate; we report how far the score moves when they change (§5).

Departures from the antecedent. Lagasio (2024) computed the index for a 2023 cross-section of 749 standalone English-language sustainability reports. Our specification departs from it in four respects.

- *Corpus and extraction.* The antecedent excluded firms that disclose ESG information inside the annual report. A balanced panel needs a document that every firm publishes every year, for all but one of our firms the annual report or the universal registration document. Because these filings embed ESG disclosure in a much larger document, we score extracted ESG passages rather than whole reports (§3.1).
- *Substance as density.* The antecedent derives substance from TF-IDF weights and states neither how they are aggregated over the vocabulary nor whether document vectors are normalised. We count vocabulary mentions per 100 tokens, because under L2 vector normalisation, the usual default, a TF-IDF substance score does not register the amount of ESG content (§3.4).
- *Dictionary sentiment.* Tone is measured with the Loughran–McDonald finance dictionary instead of the general-purpose TextBlob polarity used by the antecedent.
- *z -scores.* The antecedent prints a difference of z -scores but describes a min–max rescaling. We use z -scores as the score normalisation of every component (§3.4).

Because corpus, vocabulary and sentiment dictionary all differ, our scores are not numerically comparable with those of the antecedent, and no such comparison is attempted.

Roadmap. §2 describes the corpus and §3 the extraction, vocabulary, scores and inference. §4 reports the scores over time, against the confound of reporting mandates, by firm group and by pillar; §5 tests their sensitivity to the choices of the method, and §6 states the limitations.

2 Data

The corpus consists of 343 reports of 49 listed firms from seven euro-area countries, one report per firm and year from 2018 to 2024. The panel is balanced: every firm contributes seven reports and every year 49. Of the 344 files collected, one 2019 report was present twice, as a re-issue with identical text, and the copy was removed by comparing document contents. For each firm-year we take one recurring annual document: an annual report (235 reports), a universal registration document (URD, 101) or, for one firm, a standalone sustainability report (7). Table 1 gives the corpus by country and document type. The document-type mix barely moves over time (12 URDs in 2018, 14 or 15 in every later year). Every measure is computed on the ESG zones extracted from each report, not on whole documents (§3.1). Firms enter the paper only through their country, their industry and anonymous positions.

Each firm carries one supersector of the Industry Classification Benchmark (ICB), assigned at firm level and not derived from the reports. The 49 firms fall into 16 supersectors and 10 of the 11 ICB industries: Financials (11 firms), Consumer Discretionary (10), Industrials (9), Consumer Staples (5), Health Care and Technology (3 each), and Basic Materials, Energy, Telecommunications and Utilities (2 each); Real Estate is absent.

Table 1: Corpus composition by country and document type. Industries: number of ICB industries represented among the country’s firms. Every firm contributes one report in each year from 2018 to 2024.

Country	Firms	Industries	Annual report	URD	Sustainability report	Total
France	17	7	13	99	7	119
Germany	14	7	98	0	0	98
Netherlands	6	5	40	2	0	42
Italy	5	4	35	0	0	35
Spain	4	3	28	0	0	28
Finland	2	2	14	0	0	14
Belgium	1	1	7	0	0	7
Total	49	10	235	101	7	343

Country and document type. Document type is almost entirely determined by country: 99 of the 101 URDs are French and all 98 German reports are annual reports, which reflects national reporting practice rather than sampling. Within the pooled panel a country effect and a document-type effect can be told apart only through 15 documents (13 French annual reports and 2 Dutch URDs), and the seven standalone sustainability reports belong to a single French firm. We treat this as a property of the design: no document-type effect on either score is identified, and since type is constant within 43 of the 49 firms, firm fixed effects absorb nearly all of it (§3.5).

Industry. Industry cuts across countries: every industry has firms in at least two countries, and France and Germany each span seven of the ten, so industry contrasts can be drawn within a single country and document regime (§4.3). Industry cannot, however, be separated from the firm: industries hold 2 to 11 firms, industry is constant within each firm and so absorbed by firm fixed effects, and because the URD filers are 15 French firms and one Dutch firm, an industry’s URD share is essentially its French share.

3 Method

This section defines every quantity used later in the paper: extraction, the vocabulary, the components and scores, the reasons for three design choices, and panel inference (§3.1–3.5).

3.1 Extraction of ESG passages

The reports of §2 are annual reports and universal registration documents, in which ESG disclosure is embedded in a much larger financial and legal document. Scored whole, such a document would let financial-statement bulk and document length dominate every component. We therefore first extract its ESG passages; every later measure is computed on them alone.

Each PDF is read as text blocks in reading order, and a block becomes a paragraph only after five filters: blocks entirely within the top or bottom 8 % of the page are dropped; short margin blocks that recur, digits aside, on more than 30 % of a document’s pages are dropped as running headers or footers; tabular blocks (a high share of digits in a few lines, or short lines of which the first lacks terminal punctuation) are dropped unless they contain a vocabulary match; short all-capital blocks are taken as section headings, which label the paragraphs that follow instead of entering the text; and navigation text (tables of contents, GRI and ESRS correspondence tables, stray page references), recognised by dotted leaders, trailing page numbers, multi-level section numbers and disclosure codes, is dropped, because such lists of ESG section names would otherwise pass the density rule below. Paragraphs are then stripped of web addresses, rejoined across line-break hyphenation and collapsed to single spaces.

Each paragraph is matched against the vocabulary of §3.2 on its unlemmatised text, ignoring case, with spaces and hyphens interchangeable between the words of a phrase and longer entries matched before the shorter ones they contain, so that *carbon footprint* counts once. A paragraph with at least one match per 100 words is retained together with one paragraph on each side; overlapping or adjacent windows are merged into a single zone, and zones shorter than 150 characters are discarded. The zones of a report, in document order, form its extracted text. Across the 343 reports, extraction kept 66,171 zones holding 28.1 million of the 71.0 million words of the PDF text layers, 39.6 % (per report, median 38.3 %, range 7.1–75.6 %). The procedure depends only on the PDF and the vocabulary, and its recall is bounded by the vocabulary: a passage that uses none of its entries is never selected.

3.2 ESG vocabulary

Candidate entries came from three sources: the disclosure topics and terminology of reporting frameworks and standards (GRI, SASB, TCFD and the ESRS), including the names of the frameworks and regulatory regimes under which firms label their disclosure; the ESG vocabulary of dictionary-based studies of corporate disclosure; and a structured review of sample reports from the corpus, in which independent readers listed the ESG terms they found and independent evaluators judged the pooled list. **[Author: identify the reviewers of the candidate lists.]** A candidate was admitted if its occurrences in the extracted text, read in context, predominantly carry an ESG sense (a disclosure, target, data point or policy on an environmental, social or governance matter). Candidates from the review had in addition to be dispersed across firms, with at least 50 occurrences in the reports of at least 5 of the 49 firms. A polysemous head was admitted only inside enumerated collocations (*carbon footprint* rather than *footprint*; the ESG collocations of *social* and *water*), and a morphological family or a term carrying digits was written as a pattern (every word beginning *sustainab-* or *recycl-*; CO₂e; Scope 1–3; ISO standard numbers). Each entry belongs to one pillar: environmental (E), social (S), governance (G), or transversal (TRANS) for umbrella terms and reporting regimes (*sustainability*, *ESG*, *CSRD*). Entries whose occurrences concentrate in one industry are designated sectoral; they are part of the vocabulary, and excluding them is a sensitivity check (§5). Table 2 gives the composition.

Terms were admitted after reviewing a sample of their occurrences in context, and the frequency-weighted precision of the resulting vocabulary is estimated from those samples, not measured directly on the final extraction, at about 82%. The stability of the scores under perturbations of the vocabulary is reported in §5.

One vocabulary drives both extraction and scoring: all 434 entries select the passages, and the 402 terms, lemmatised, are what the substance component counts. Changing the vocabulary therefore changes the text on which every component is computed, not only the substance count. For scoring, the extracted text is lowercased, tokenised and lemmatised with a general-purpose English language model; stopwords (the standard list and 189 corpus-specific words), punctuation, numerals and tokens shorter than three characters are removed, and the vocabulary passes through the same processing, so that each entry is matched in the representation of the text it is counted on. Eleven phrases that this processing would empty or reduce to a generic word (*well-being*, both of whose words are stopwords; *say on pay* to *pay*; *ISO 14001* to *iso*) are first replaced by single protected tokens, in text and vocabulary alike. The 402 terms yield $V = 399$ distinct lemmatised scoring entries.

3.3 Components and scores

Let $d = 1, \dots, n$ index the $n = 343$ reports and $j = 1, \dots, V$ the scoring entries. Each report has two representations of its extracted text: the *lemmatised text* of §3.2, with L_d tokens, and the *raw text*, with case, digits and stopwords intact, with R_d tokens. Let c_{dj} be the number of occurrences of entry j in the lemmatised text of d , counted over word sequences of one to four tokens (the length of the longest entry). Entries are counted independently, so an occurrence of a multi-word entry also counts for any shorter entry it contains.

Substance. SUS is the density of ESG mentions, the number of vocabulary occurrences per 100 lemmatised tokens, with uniform term weights:

$$\text{SUS}_d = \text{SUS}_d^{\text{dens}} = 100 \cdot \frac{\sum_j c_{dj}}{L_d},$$

written $\text{SUS}_d^{\text{dens}}$ where it is compared with the alternatives of §3.4. It uses no document frequencies, so a report’s value does not depend on the other reports in the corpus.

Tone. SEN is the Loughran–McDonald polarity (Loughran and McDonald, 2011) of the lemmatised text,

$$\text{SEN}_d = \frac{s_d^+ - s_d^-}{s_d^+ + s_d^- + \varepsilon}, \quad \varepsilon = 10^{-6},$$

Table 2: Composition of the ESG vocabulary. Terms and collocations are literal phrases; a collocation stands in for a polysemous head. Sectoral entries cut across the three type columns. Scoring entries are the distinct lemmatised terms (patterns act at extraction only); the last column is their share of the 1,047,951 vocabulary occurrences in the extracted text of the 343 reports.

Pillar	All entries					Scoring entries	
	Terms	Colloc.	Patterns	Total	Sectoral	Entries	Occ. (%)
E (environmental)	123	29	16	168	14	152	38.2
S (social)	105	23	11	139	14	127	17.6
G (governance)	65	27	3	95	8	90	30.0
TRANS (transversal)	27	3	2	32	0	30	14.2
Total	320	82	32	434	36	399	100.0

where s_d^+ and s_d^- count the tokens whose stem is in the dictionary’s positive and negative lists, and ε only prevents division by zero. It is a bounded ratio on $[-1, 1]$ that depends on the balance of positive and negative words, not on how many there are.

Quantification. QUANT is the number of pattern matches per raw token, summed over four families: percentages; numbers of four or more digits other than years; physical units (tonnes, CO₂e, GHG, energy, power and volume units); and the names of reporting frameworks (GRI, TCFD, SASB, CSRD, SDG, the taxonomy and others). It is computed on the raw text because preprocessing removes numerals.

Hedging. HEDGE is the share of the words of the raw text (letter runs, hyphenated compounds kept whole) that belong to the Uncertainty, Weak Modal, Strong Modal and Constraining lists of the Loughran–McDonald master dictionary, 495 word forms taken as written. It is computed on the raw text because the modal verbs (*may, might, could, must, will*) are stopwords. Matching is by form, not by sense: the five most frequent forms (*risk, risks, will, may, requirements*) make up 49.4 % of all matches, so HEDGE registers to a large extent how often a report names risk.

Breadth and pillars. BREADTH_d = $\exp(-\sum_j p_{dj} \ln p_{dj})$, with $p_{dj} = c_{dj} / \sum_k c_{dk}$, is the effective number of distinct entries a report uses, a Hill number of order one (Hill, 1973); it is a diagnostic, not a component of either score. The pillar density $100 \sum_{j \in p} c_{dj} / L_d$ counts the mentions of pillar-*p* entries per 100 lemmatised tokens, and the four pillar densities sum to SUS_d; for a set of reports, the pillar share pools the mentions of the set. Both are descriptive and enter neither score.

Scores. Each component x is rescaled across the corpus by a *z*-score, $Z(x)_d = (x_d - \bar{x}) / \sigma_x$, with $\bar{x}$ the mean and σ_x the population standard deviation over all 343 reports pooled across the seven years, so that yearly means of a score are comparable. The two scores are

$$\text{ESGSI}_d = Z(\text{SEN})_d - Z(\text{SUS})_d, \quad (1)$$

$$\text{ESGSI}_d^{\text{ext}} = Z(\text{SEN})_d - Z(\text{SUS})_d - w_q Z(\text{QUANT})_d + w_h Z(\text{HEDGE})_d, \quad w_q = w_h = 0.5. \quad (2)$$

ESGSI (1) is the main score of the paper. It keeps the architecture of Lagasio (2024), tone minus substance with each rescaled across the sample, with substance measured by density and tone by a finance-specific dictionary in place of a general-purpose polarity score. The Extended ESGSI (2) is the new metric this paper proposes. Quantified disclosure is further evidence of substance, so $Z(\text{QUANT})$ is subtracted, on the same side as $Z(\text{SUS})$; hedging qualifies whatever a report asserts, so $Z(\text{HEDGE})$ is added, on the side of tone. The weights give each added component half the weight of an original one, so that together they weigh as much as one of them and the tone–substance contrast remains the core of the score. They are a stated choice, not an estimate, since no external criterion is available to calibrate them; §5 reports the scores over a grid of weights.

Both scores are continuous and corpus-relative: each has mean zero by construction, and its sign says only whether a report lies above or below the corpus mean of the combination. We compare positions within the corpus: differences of means, trends, rank correlations and the overlap of the top and bottom deciles.

3.4 Why density, *z*-scores and no threshold

Lagasio (2024) computes substance with a TF-IDF vectoriser (Pedregosa et al., 2011) over an author-built keyword list and discloses four of its settings: two document-frequency bounds, a vocabulary cap and the range of word sequences. It does not state whether each document’s

term vector is scaled to unit Euclidean length (vector normalisation), nor how the per-term values become a document score. The aggregation is immaterial once scores are standardised, since a sum and a mean over the vocabulary differ by the factor $1/V$. Vector normalisation is the vectoriser’s default and all four disclosed settings depart from defaults; inferring, as a reading and not a finding, that only departures were reported, the closest reading of the antecedent is $\text{SUS}^{\text{tfidf-}L2}$ below. We also compute $\text{SUS}^{\text{tfidf-}\ell}$, which differs from SUS^{dens} only in the idf weights:

$$\text{SUS}_d^{\text{tfidf-}L2} = \frac{1}{V} \sum_j \frac{c_{dj} \text{idf}_j}{\|\mathbf{c}_d \odot \mathbf{idf}\|_2}, \quad \text{SUS}_d^{\text{tfidf-}\ell} = 100 \cdot \frac{\sum_j c_{dj} \text{idf}_j}{L_d},$$

where $\text{idf}_j = \ln[(1+n)/(1+df_j)] + 1$ and df_j is the number of reports containing entry j .

A term vector and any positive multiple of it have the same unit vector, so $\text{SUS}^{\text{tfidf-}L2}$ keeps the mix of terms and discards their amount, a property suited to retrieval (Spärck Jones, 1972; Manning et al., 2008) but not to a score that sets tone against the amount of substance behind it. Panel A of Table 3 shows this on the report at the median of SUS^{dens} : multiplying its ESG counts by up to 20 raises its density from 6.78 to 33.79 and leaves $\text{SUS}^{\text{tfidf-}L2}$ unchanged, and diluting it with 152,378 non-ESG tokens divides its density by 3.5 and leaves $\text{SUS}^{\text{tfidf-}L2}$ identical. What remains after vector normalisation is dispersion over terms: the sum of a non-negative unit vector lies between 1 and $\sqrt{V}$ and grows as mentions spread over more entries. On our corpus $\text{SUS}^{\text{tfidf-}L2}$ correlates 0.955 with BREADTH and 0.405 with SUS^{dens} ; under vector normalisation, the substance component measures topical breadth, not the amount of disclosure.

An empirical check needs a yardstick built by other means. QUANT counts pattern matches on the raw text, with no vocabulary, vectoriser or document frequencies, and a report that discloses substance should also quantify it (Campbell and Fiske, 1959). On the full bases (Table 3, panel B) density correlates 0.63 with QUANT and $\text{SUS}^{\text{tfidf-}L2}$ 0.30, and the firm-bootstrap interval of the difference excludes zero. The two instruments share 17 entries (framework acronyms,

Table 3: Why density. Panel A: a synthetic document written out from the ESG counts of the report at the median of SUS^{dens} , with every count multiplied by k , and the real report with 152,378 non-ESG tokens appended; all 343 reports re-vectorised in every run; $\text{SUS}^{\text{tfidf-}L2}$ relative to the first row of each block (maximum absolute deviation 3.5×10^{-18}). Panel B: Pearson correlation with QUANT ($n = 343$, 49 firms); disjoint bases remove from the vocabulary the 17 entries also matched by a QUANT pattern (399 to 382 entries) and restrict QUANT to percentages and large numbers; 95 % percentile intervals from 9,999 bootstrap samples of firms, each with its seven reports.

<i>A. Amount of ESG content and the substance specifications</i>				
	Tokens	ESG mentions	SUS^{dens}	$\text{SUS}^{\text{tfidf-}L2}$ (rel.)
Amplified, $k = 1$	60,951	4,131	6.78	1.000000
Amplified, $k = 2$	70,612	8,262	11.70	1.000000
Amplified, $k = 5$	99,595	20,655	20.74	1.000000
Amplified, $k = 20$	244,510	82,620	33.79	1.000000
Real report	60,951	3,984	6.54	1.000000
Diluted	213,329	3,984	1.87	1.000000
<i>B. Convergent validity: correlation with QUANT</i>				
	Full bases		Disjoint bases	
	r	95 % interval	r	95 % interval
SUS^{dens}	0.6305	[0.523, 0.732]	0.3992	[0.227, 0.539]
$\text{SUS}^{\text{tfidf-}\ell}$	0.6392	[0.550, 0.736]	0.4007	[0.256, 0.535]
$\text{SUS}^{\text{tfidf-}L2}$	0.2984	[0.041, 0.551]	0.1850	[−0.072, 0.483]
$\text{SUS}^{\text{dens}} - \text{SUS}^{\text{tfidf-}L2}$	0.3321	[0.075, 0.589]	0.2142	[−0.143, 0.519]

the taxonomy family, CO₂, GHG, ISO standards) and read the same zones, so we repeat the test on disjoint bases. Every correlation falls, by 37–38 %, and the ratio between the two specifications does not (2.16 against 2.11), but the interval of the difference now includes zero: convergent validity corroborates a density specification without carrying the choice on its own, which rests on the invariance above. The two density specifications are not separated by this test (difference 0.009, interval $[-0.018, 0.044]$ on the full bases); we adopt SUS^{dens} because it estimates nothing on the sample.

The antecedent prints its index as a difference of z -scores but states that the components are min–max rescaled, and its published statistics are min–max: its mean index, -0.2179 , is the difference of its component means, $0.3727 - 0.5906$ (Lagasio, 2024, p. 6), whereas a difference of z -scores has mean zero. Both score normalisations are increasing affine maps and preserve ranks within a component; once two components are differenced, they differ in how they equalise spreads and where they put zero. On our corpus ranks barely move (Spearman correlation 0.997 between the z -score and min–max versions of ESGSI), but location does: under min–max the mean of the score depends on where each component’s mass sits within its observed range ($+0.004$ with SUS^{dens}, -0.203 with SUS^{tfidf-L2}, on the same reports), and a single extreme report can shift it. A z -score puts each component’s zero at its corpus mean and measures it in its own standard deviations, so a score’s sign and size describe a position in the corpus and nothing else. The antecedent also assigns every report to one of two classes by the sign of its index. We apply no threshold of any kind. Zero is the corpus mean of the combination, so a cut-off on a difference of standardised components describes the relative location of two distributions, not the conduct of a firm, and no external benchmark exists against which a boundary could be calibrated for these reports (§6).

3.5 Inference on the panel

Write $d = (i, t)$ for the report of firm $i = 1, \dots, M$ in year t , with $M = 49$ in the balanced panel of §2. Trends are estimated with firm fixed effects, regressing the firm-demeaned outcome on the firm-demeaned year; in the balanced panel the slope equals the mean of the firm-level least-squares slopes. Standard errors are clustered by firm with the usual small-sample correction, and t statistics are referred to a t distribution with $M - 1$ degrees of freedom, M being the number of firms in the sample or subsample. Clustering allows any correlation among a firm’s seven reports but not a shock common to all firms in one year, which with seven years cannot be clustered on; the slope summarises the yearly means. Because 49 clusters are few, the fixed-effects slope is also tested with a restricted wild cluster bootstrap (Cameron et al., 2008): under a zero slope the residuals are the demeaned outcomes, each of $B = 9,999$ replications multiplies all residuals of a firm by one Rademacher weight (± 1 with equal probability), and the p -value is the share of replications whose cluster-robust t is at least as large in absolute value as the observed one. Slopes by industry or country interact the demeaned year with group indicators. Each equals the mean firm slope of its group, but for a group of M_k firms its clustered variance rests on $M_k - 1$ free contributions and is too narrow for small groups, so we read group slopes through the t interval on the M_k firm slopes ($M_k - 1$ degrees of freedom). Equality across groups is tested with a Wald test on the clustered variance, anti-conservative when a group has a single firm. Yearly means carry 95 % intervals from a regression on year indicators, clustered by firm.

4 Results

This section reports ESGSI (1) and ESGSI^{ext} (2) on the panel of §2, in corpus standard deviations, with trends and intervals estimated as in §3.5. Industries are ordered by mean ESGSI throughout.

4.1 The yearly series and the trend

Both scores fall in every year of the window: mean ESGSI from 0.502 in 2018 to -0.610 in 2024, mean $\text{ESGSI}^{\text{ext}}$ from 0.696 to -0.802 (Table 4, Figure 1a). The slope is -0.181 per year for ESGSI (95% interval $[-0.250, -0.111]$, $p_{\text{cl}} = 3.7 \times 10^{-6}$) and -0.260 for $\text{ESGSI}^{\text{ext}}$ ($[-0.340, -0.180]$, $p_{\text{cl}} = 3.7 \times 10^{-8}$); for neither did any of the 9,999 bootstrap replications reach the observed statistic. The yearly intervals nonetheless overlap widely, and three reading rules apply: the slope summarises an uneven path, not a constant rate; the zero crossing between 2020 and 2021 is not an event, since both scores average zero over the window by construction; and the clustered errors do not cover a shock common to all firms in one year (§3.5).

What moves is substance. By (1), a decline means tone falling or substance rising, and it is the second (Figure 1b): SUS^{dens} rises in every year, from 5.75 to 8.05 mentions per 100 tokens, because mentions per report grow faster (+205%) than the extracted text that divides them (+119%). Tone does not move: SEN has no detectable trend ($p_{\text{cl}} = 0.581$, $p_{\text{wild}} = 0.577$), and its yearly means dip in 2020 and rise in 2024, against the decline; a failure to reject is not a demonstrated null. $\text{ESGSI}^{\text{ext}}$ falls faster because of quantification: QUANT rises by about half over the window, though not in every year, while HEDGE is flat, its standardised yearly means within ± 0.06 of zero.

The panel is balanced, so the series is not a change in the set of firms or in the country mix, and the document-type mix barely moves (§2). Within Germany, fourteen firms with one country and one document type, both slopes are steeper than the pooled ones: -0.268 for ESGSI ($p_{\text{cl}} = 0.002$, $p_{\text{wild}} = 0.003$, 13 degrees of freedom) and -0.376 for $\text{ESGSI}^{\text{ext}}$ ($p_{\text{cl}} = 4.3 \times 10^{-5}$, $p_{\text{wild}} = 3.0 \times 10^{-4}$). A trend of this sign therefore exists where the country/document-type confound is absent, which does not show that confound absent from the pooled slope.

4.2 Regime vocabulary and the mandate confound

Over 2018–2024 European sustainability reporting moved from largely voluntary to substantially prescribed. A firm that adds a taxonomy-alignment table or a standards-referenced sustainability statement adds ESG vocabulary without necessarily changing anything it does, and SUS^{dens} cannot tell the two apart. The vocabulary names regimes that did not exist or were not in force in 2018, and the transversal pillar that holds them nearly doubles its share of all mentions, from

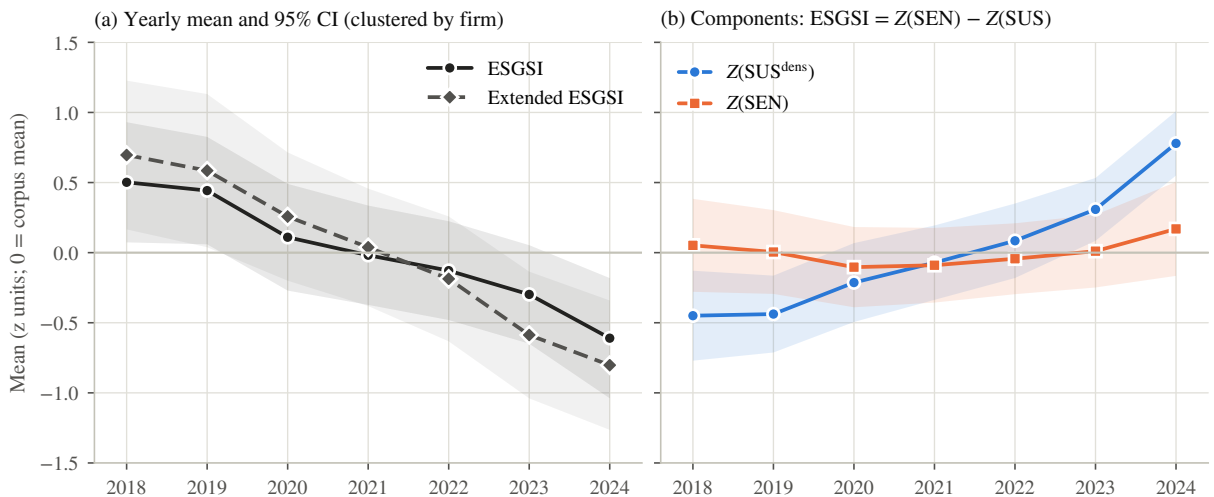


Figure 1: (a) Mean ESGSI and $\text{ESGSI}^{\text{ext}}$ by year, 49 reports per year, with 95% intervals from year indicators and standard errors clustered by firm. (b) The same for $Z(\text{SUS}^{\text{dens}})$ and $Z(\text{SEN})$, whose difference is ESGSI (1). All values in corpus standard deviations; zero is the mean over the 343 reports.

Table 4: Yearly means and trend regressions; 343 reports of 49 firms, 49 per year. Panel A: means of the two scores and of their standardised components (1), (2); z -scores are taken over the pooled corpus (§3.3), so every row averages zero over the window. Panel B: slope on calendar year with firm fixed effects; standard errors and 95% intervals clustered by firm (t_{48}); p_{wild} from a restricted wild cluster bootstrap with 9,999 replications (§3.5); R_w^2 within firm. Component slopes are in their own units: SUS^{dens} in mentions per 100 tokens, QUANT and HEDGE in units of 10^{-4} , all per year.

<i>A. Yearly means</i>							
	2018	2019	2020	2021	2022	2023	2024
ESGSI	0.502	0.443	0.110	-0.018	-0.128	-0.298	-0.610
ESGSI ^{ext}	0.696	0.585	0.258	0.038	-0.187	-0.587	-0.802
Z(SEN)	0.052	0.005	-0.103	-0.090	-0.043	0.011	0.169
Z(SUS ^{dens})	-0.450	-0.438	-0.213	-0.072	0.085	0.308	0.779
Z(QUANT)	-0.402	-0.245	-0.276	-0.073	0.123	0.522	0.351
Z(HEDGE)	-0.012	0.040	0.019	0.039	0.005	-0.057	-0.034
<i>B. Trend</i>							
	Slope	SE _{cl}	95% interval		p_{cl}	p_{wild}	R_w^2
ESGSI	-0.181	0.035	-0.250	-0.111	3.7×10^{-6}	$< 10^{-4}$	0.211
ESGSI ^{ext}	-0.260	0.040	-0.340	-0.180	3.7×10^{-8}	$< 10^{-4}$	0.295
SEN	0.0020	0.0037	-0.0053	0.0094	0.581	0.577	0.003
SUS ^{dens}	0.366	0.041	0.283	0.449	1.1×10^{-11}	$< 10^{-4}$	0.364
QUANT	4.7	0.9	2.9	6.5	3.4×10^{-6}	$< 10^{-4}$	0.206
HEDGE	-0.4	0.8	-1.9	1.2	0.633	0.634	0.002

10% to 19% (§4.4).

The purely definitional part can be bounded by deleting entries and rescoreing, with the extraction held fixed. Deleting an explicit list of 22 regime, standard and rating names (3.8% of all mentions; deliberately over-inclusive, since several predate the window) leaves an ESGSI slope of -0.149 ($p_{\text{cl}} = 8.9 \times 10^{-5}$, $p_{\text{wild}} = 1.0 \times 10^{-4}$), about four fifths of the original, with a rank correlation of 0.992 with the full-vocabulary score. Deleting the whole transversal pillar (30 scoring entries holding 21 of the 22 names and 14.2% of all mentions) leaves -0.117 ($p_{\text{cl}} = 0.002$, $p_{\text{wild}} = 0.003$), about two thirds, with $\rho = 0.973$. ESGSI^{ext} keeps -0.228 and -0.196 of its -0.260 . The trend is not an artefact of counting regime names.

Neither check disposes of the mandate confound. A regime elicits substantive disclosure as well as its own names (emissions, transition plans, most of the environmental vocabulary) in the words a firm that had changed its conduct would also use, and the environmental pillar is where the rise sits (§4.4). A mandate is a common time shock, collinear with calendar year; the balanced panel, the German subsample and the fixed effects remove cross-sectional composition, not anything that moves with time. Identification would need variation in exposure, which 49 large issuers, all in scope, do not provide, so the confound is left unresolved. One consequence belongs to the architecture: since both scores fall when substance rises, a firm that adds a mandated disclosure moves down the scale for having complied.

4.3 Firms, industries and countries

Most of what the scores record is a difference between firms. For ESGSI, 66.7% of the total sum of squares lies between firm means and 7.2% between year means, and the intraclass correlation of a firm’s seven reports, net of year, is 0.676 (ESGSI^{ext}: 68.5%, 9.4%, 0.720). Positions persist from one year to the next and reshuffle over the window: the Spearman correlation of firms’ ESGSI between consecutive years averages 0.849 (range 0.728 to 0.915) but is 0.456 between 2018 and 2024 (ESGSI^{ext}: 0.823 and 0.393). The decline is general but not uniform: 39 of the

49 firm slopes are negative for ESGSI and 43 for ESGSI^{ext}.

Industry and country are constant within firm, so every comparison between groups rests on 49 firms in cells of one to seventeen (Table 5), and every country contrast is also a document-type contrast (§2).

Levels. Industry means of ESGSI run from 1.125 (Technology, three firms) to -1.919 (Energy, two); Utilities has by far the highest SUS^{dens} and the second-lowest score. Both ends rest on two or three firms, dispersion within an industry can exceed the distance between industries (Financials’ firm means span -1.41 to 2.09), and the ordering moves within the window (Figure 2): Basic Materials and Telecommunications go from well above the corpus mean in 2018 to below it in 2024. ESGSI^{ext} reorders the industries where quantification and hedging differ most: Financials, with the highest mean HEDGE and among the lowest QUANT, moves from 0.041 to 0.791 , and Utilities, with the highest QUANT, from -1.171 to -2.573 . Within a single country and document regime, industry does not order firms reliably: it accounts for an adjusted R^2 of 0.191 of the variance of ESGSI firm means in France (17 firms, seven industries) and of -0.464 in Germany (14 firms, seven industries).

Germany stands apart among countries (1.003 against -0.335 for France), and in each of the five industries present in both countries the German mean exceeds the French one: the gap survives holding industry fixed, and so does the document-type contrast it carries. By document type, annual reports average 0.167 , URDs -0.263 and the one firm’s sustainability reports -1.792 ; neither a country nor a document-type effect is identified. At report level, industry and country each add more to the explained variance of ESGSI than year does, yet firm identity still adds 0.301 of R^2 after year, industry, country and document type (ESGSI^{ext}: 0.259); across the 49 firm means, industry and country together reach an adjusted R^2 of 0.287 (ESGSI^{ext}: 0.421). Most of the difference between firms is firm-specific.

Table 5: Scores by ICB industry and by country, each ordered by mean ESGSI. Firms: number of firms, each contributing seven reports. Means are over the group’s reports; SUS^{dens} in mentions per 100 tokens. Slope and 95% interval: group slope and t interval on the group’s firm-level slopes (§3.5); none for a one-firm group.

	Firms	SUS ^{dens}	ESGSI			ESGSI ^{ext}		
			Mean	Slope	95% interval	Mean	Slope	95% interval
<i>Industry (ICB)</i>								
Technology	3	6.17	1.125	−0.328	[−0.827, 0.171]	1.268	−0.399	[−1.136, 0.338]
Telecommunications	2	5.89	0.935	−0.404	[−1.816, 1.007]	0.750	−0.495	[−1.882, 0.892]
Industrials	9	6.41	0.297	−0.137	[−0.280, 0.006]	0.200	−0.276	[−0.365, −0.188]
Consumer Discretionary	10	6.56	0.138	−0.287	[−0.471, −0.103]	0.078	−0.334	[−0.539, −0.128]
Financials	11	5.92	0.041	−0.097	[−0.221, 0.027]	0.791	−0.244	[−0.383, −0.104]
Health Care	3	6.67	−0.104	0.066	[−0.819, 0.952]	0.013	−0.019	[−0.892, 0.855]
Basic Materials	2	7.08	−0.190	−0.437	[−1.236, 0.361]	−0.731	−0.448	[−1.999, 1.103]
Consumer Staples	5	7.17	−0.575	−0.239	[−0.374, −0.104]	−0.946	−0.240	[−0.382, −0.098]
Utilities	2	9.86	−1.171	0.282	[−0.933, 1.497]	−2.573	0.499	[−3.595, 4.594]
Energy	2	7.33	−1.919	−0.286	[−0.953, 0.381]	−2.642	−0.455	[−0.716, −0.195]
<i>Country</i>								
Germany	14	5.60	1.003	−0.268	[−0.419, −0.116]	1.107	−0.376	[−0.511, −0.241]
Belgium	1	6.37	−0.203	−0.372	—	−0.043	−0.401	—
Finland	2	6.26	−0.242	−0.123	[−2.287, 2.041]	0.216	−0.399	[−0.565, −0.233]
France	17	6.84	−0.335	−0.182	[−0.296, −0.068]	−0.575	−0.250	[−0.367, −0.132]
Spain	4	8.51	−0.361	−0.055	[−0.336, 0.226]	−0.460	−0.074	[−0.361, 0.214]
Netherlands	6	7.04	−0.363	−0.181	[−0.446, 0.085]	−0.031	−0.302	[−0.570, −0.034]
Italy	5	6.65	−0.808	−0.017	[−0.360, 0.325]	−0.817	0.013	[−0.602, 0.628]



Figure 2: Mean (a) ESGSI and (b) $\text{ESGSI}^{\text{ext}}$ by ICB industry and year, in corpus standard deviations, on a diverging colour scale centred on the corpus mean. Industries are ordered by mean ESGSI; the number of firms, in parentheses, is also the number of reports averaged in each cell.

Slopes. Group slopes are read through the intervals on the group's firm-level slopes (Table 5, §3.5). Utilities shows why: its positive ESGSI slope has a clustered interval of $[0.143, 0.421]$ and a firm-slope interval of $[-0.933, 1.497]$. Read this way, the ESGSI decline is distinguishable from zero in Consumer Discretionary (10 firms), Consumer Staples (5), Germany (14) and France (17); Financials (11) and Industrials (9) have the flattest declines among the larger industries, with intervals that just include zero. Under $\text{ESGSI}^{\text{ext}}$, Financials, Industrials and the Netherlands also decline distinguishably. No group of one to three firms, including the two industries with a positive slope, supports a statement about its ESGSI slope.

4.4 Pillar composition

The rise in mentions is not spread evenly across the pillars of §3.2 (Figure 3b). The environmental share of all mentions grows from 32% in 2018 to 42% in 2024 and the transversal share from 10% to 19%, while governance falls from 39% to 23% and the social share stays between 16% and 20%. In density, environmental mentions rise in every year, from 1.73 to 3.25 per 100 tokens, and transversal ones from 0.63 to 1.52, while governance mentions fall from 2.34 to 1.90: governance loses ground absolutely, not only by dilution.

Across industries the composition follows the business (Figure 3a). The environmental share runs from 26% (Telecommunications) to 63% (Utilities), whose environmental density of 6.04 mentions per 100 tokens, against 4.11 in Energy, the next industry, gives it the highest SUS^{dens} in Table 5; Telecommunications and Financials are the governance-heavy industries (39% and 38%). The entries most over-represented in an industry are its own vocabulary (*hydroelectric* and *smart meter* in Utilities, *oil spill* and *methane* in Energy), so a firm scores substance partly for describing its business. The sectoral entries hold 1.0% of all mentions and at most 2.7% in

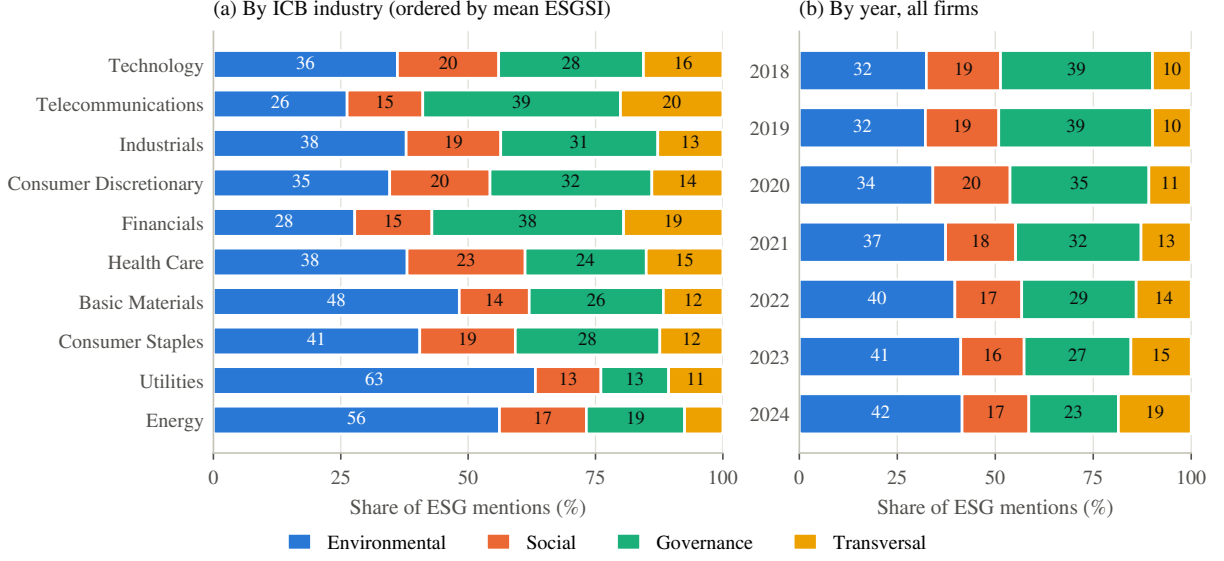


Figure 3: Share of all vocabulary mentions falling in each pillar, pooled over the group’s reports, in percent: (a) by ICB industry, ordered by mean ESGSI; (b) by year, all 49 firms. The unlabelled segment (Energy, transversal) is 8%.

any industry (Consumer Staples), too little to account for the position of Utilities or Energy. Mentions are concentrated (15 of the 399 scoring entries hold 48.1% of them); scoring without the sectoral entries, without each pillar and without the most frequent entry is examined in §5.

What the results license. Neither score has cardinal meaning (§3.3): a yearly mean of -0.610 is a position within this corpus, not a quantity of anything, and no level here is comparable with another application of the architecture. What is interpretable is the ordering within the sample and the direction of change: both scores decline over 2018–2024 because ESG vocabulary per unit of extracted text rose while tone did not, and firms differ far more than years. None of this is evidence about conduct: the reporting mandates of the same window could move the scores with conduct held fixed, and this design cannot exclude them (§4.2).

5 Robustness

We vary three choices of the method—the substance specification, the weights of the Extended ESGSI and the vocabulary—and compare each variant with a reference score over the same 343 reports. Every comparison uses three measures: the Spearman correlation ρ with the reference; the overlap of the extreme deciles, that is, how many of the 34 reports with the highest (lowest) reference score remain among the 34 highest (lowest), written top/bottom; and the within-firm slope per year, estimated with firm fixed effects and firm-clustered standard errors (§3.5). The reference slopes are -0.181 for ESGSI and -0.260 for $\text{ESGSI}^{\text{ext}}$ (§4.1).

Substance specification. Computing ESGSI with the length-normalised TF-IDF measure $\text{SUS}^{\text{tfidf-}\ell}$ instead of SUS^{dens} leaves its ranking almost unchanged ($\rho = 0.993$; 33/32 of the extreme deciles kept). With the L2 vector-normalised measure $\text{SUS}^{\text{tfidf-L2}}$ the ranking changes markedly ($\rho = 0.642$; 18/21) and the slope weakens to -0.143 . Term weighting is thus a second-order choice and vector normalisation a first-order one (§3.4).

Weights of the Extended ESGSI. Over the grid $w_q, w_h \in \{0, 0.25, 0.5, 0.75, 1\}$ of Eq. (2), ρ with the score at the default weights is at least 0.865, the minimum being at $w_q = w_h = 0$,

Table 6: Deletions from the vocabulary under simulated re-extraction ($n = 343$), against the full-vocabulary score as recomputed by the simulation (slope -0.180 for ESGSI, against -0.181 for the adopted score; -0.260 for ESGSI^{ext}). Entries: of the 434. Text: share of extracted words in paragraphs that still qualify (%). Top/bot: reports of the reference top/bottom 34 that remain. Slope: within-firm slope per year. Random rows: medians over 1,000 draws (text: mean). With caveat: six frequent entries admitted although part of their occurrences is navigation text, legal boilerplate or a non-ESG sense.

Deleted	Entries	Text	ESGSI			ESGSI ^{ext}		
			ρ	Top/bot	Slope	ρ	Top/bot	Slope
Sectoral entries	36	99.6	0.9995	33/33	-0.176	0.9994	33/32	-0.256
Pillar E	168	81.1	0.8844	28/23	-0.105	0.9088	30/26	-0.211
Pillar S	139	90.1	0.9836	32/30	-0.168	0.9886	31/31	-0.253
Pillar G	95	68.6	0.9439	28/30	-0.206	0.9622	27/29	-0.250
Pillar TRANS	32	94.8	0.9822	31/32	-0.132	0.9832	32/32	-0.205
<i>Board of directors</i>	1	93.6	0.9953	33/31	-0.188	0.9965	33/33	-0.268
With caveat	6	95.2	0.9912	32/32	-0.180	0.9940	30/31	-0.255
Random 10 %	43	96.3	0.9931	33/32	-0.174	0.9954	33/32	-0.253
Random 20 %	87	91.8	0.9820	31/31	-0.165	0.9873	31/31	-0.242
Random 30 %	130	86.6	0.9674	30/30	-0.154	0.9764	30/30	-0.231
Random 40 %	174	80.8	0.9497	29/28	-0.139	0.9611	29/29	-0.214
Random 50 %	217	74.0	0.9242	27/26	-0.118	0.9396	28/28	-0.192

where the score is ESGSI, and at least 0.973 within 0.25 of the default on each axis, where at least 28 of the top and 29 of the bottom 34 reports remain. The ranking depends mostly on w_h and the slope mostly on w_q (from -0.185 at $w_q = 0$ to -0.335 at $w_q = 1$, with $w_h = 0.5$); the slope is negative in all 25 pairs, and every firm-clustered 95 % interval excludes zero.

Vocabulary. The vocabulary selects the text as well as scoring it (§3.2), so each deletion of entries is run in two ways: with the extraction held fixed, so that only the list counted by SUS changes, and with a simulated re-extraction, in which the extraction rule of §3.1 is re-applied without the deleted entries and all components are recomputed on the paragraphs that still qualify. A smaller vocabulary can only select part of the adopted extraction, so the simulation starts from the extracted paragraphs; with the full vocabulary it reproduces the adopted scores ($\rho > 0.9999$). We delete designed sets of entries and, 1,000 times at each size, random sets of 10–50 % of the 434 entries, and read each designed deletion against 1,000 random deletions of the same size. Table 6 gives the re-extraction results for both scores, and Figure 4 those for ESGSI in both modes.

Removing the 36 sectoral entries barely moves either score ($\rho \geq 0.998$ in both modes): they carry about 1 % of the vocabulary’s matches. The environmental pillar moves the score most. Without it $\rho = 0.884$, only 23 of the bottom 34 reports remain, and the slope falls to -0.105 (SE 0.039, firm-clustered $p = 0.009$, wild cluster bootstrap $p = 0.011$). Its ρ is lower than in 998 of the 1,000 random deletions of the same size, although its entries carry no more of the matches than random sets of that size (38.2 % against 38.0 %): the environmental entries order the reports differently from the rest of the vocabulary. It is the only designed deletion whose ρ falls below the median for a random half of the vocabulary (Figure 4a). With the extraction held fixed its effect is larger ($\rho = 0.776$, 12 of the bottom 34 kept, slope -0.080 with firm-clustered $p = 0.027$), because the paragraphs extracted for environmental entries stay in the text, unscored, and dilute density. Without the transversal pillar the slope is -0.132 ; the remaining designed deletions leave it between -0.168 and -0.206 , the steepest being without pillar G, which also removes nearly a third of the extracted text. ESGSI^{ext} moves less than ESGSI, in ρ , under every

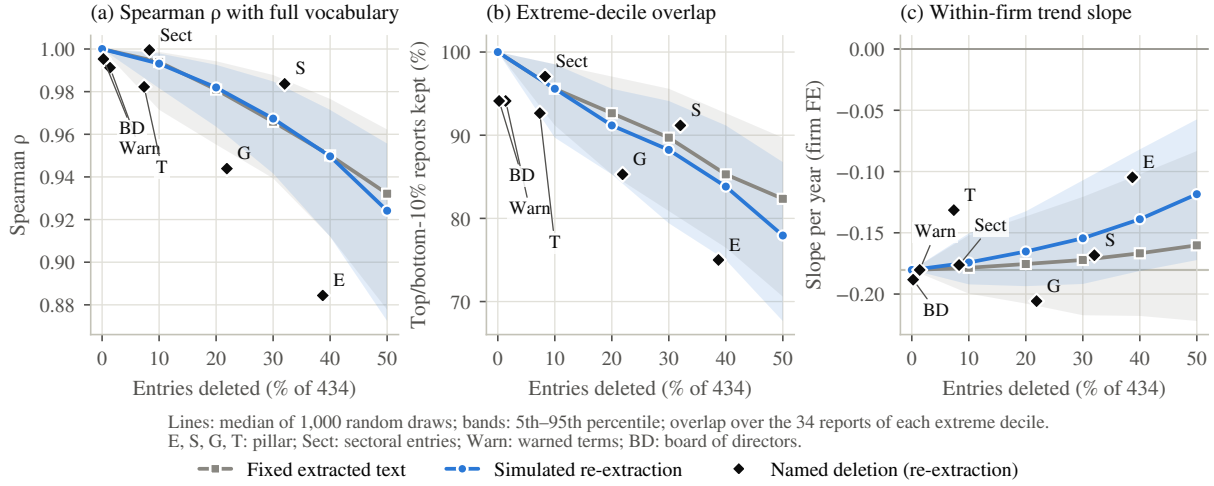


Figure 4: Stability of ESGSI under deletion of vocabulary entries ($n = 343$): random deletions in both modes, and the designed deletions of Table 6 under re-extraction, placed at the share of the 434 entries they remove (T: pillar TRANS; Warn: entries with a caveat). (a) Spearman ρ with the full-vocabulary score. (b) Mean of the top- and bottom-decile overlaps. (c) Within-firm slope per year; the horizontal line is the full-vocabulary slope.

pillar deletion.

Under random deletion the ranking degrades smoothly: with half of the entries removed the median ρ is 0.924 for ESGSI and 0.940 for ESGSI^{ext}, and 27 of the top and 26 of the bottom 34 reports remain. The slope is the least stable measure. Under re-extraction its median drifts towards zero as more is deleted, from -0.174 at 10 % to -0.118 at 50 %, because less of the original text remains; with the extraction held fixed it stays between -0.179 and -0.160 . Its sign does not change: over the 5,000 random draws the ESGSI slope is negative in all but one (a 50 % deletion under re-extraction), and the ESGSI^{ext} slope in all, in both modes.

These checks bound how far the scores depend on the choices varied; they do not validate the scores (§6).

6 Limitations

- *No external criterion.* Both scores are positions within this corpus and would move if the panel changed. Neither is tested against an external criterion (regulatory findings, controversy data, hand-coded cases), so no value is evidence about a firm's conduct; the one empirical check on substance, convergent validity with QUANT (§3.4), is internal and reads the same extracted text.
- *A vocabulary built on this corpus.* Candidates, admission decisions, pillars and sectoral entries were settled on the 343 reports that are scored, with no part held out, and admission rests on the judgement of the reviewers of the candidate lists [Author: identify the reviewers.]. Its precision is estimated rather than measured on the final extraction (§3.2), and its recall is unknown. The stability of §5 is stability around this fit; it covers deletions of entries, not additions, and the size of the trend depends on the environmental and transversal entries.
- *Country, document type and industry.* Document type is almost determined by country (§2), so neither effect is identified on its own. Industries hold 2 to 11 firms and supersectors 1 to 7, both constant within firm, so group contrasts rest on few firms (§4.3).
- *Inference.* Standard errors are clustered on 49 firms and checked with a wild cluster bootstrap, but they do not cover shocks common to all firms in a year, and seven years are

too few to cluster on (§3.5).

- *Mandate*. European sustainability reporting became progressively mandated over the window, an exposure common to all firms and collinear with calendar time, so the rise in substance cannot be split into changed conduct and compliance, and compliance itself lowers both scores (§4.2).
- *Weights*. The weights of the Extended ESGSI are a choice. Its ranking is stable near them, but the size of its trend scales with w_q (§5).
- *Dictionaries*. SEN applies a finance dictionary to non-financial narrative, without negation handling, and as a bounded ratio it ignores how many sentiment words a text contains. HEDGE matches by form (§3.3): *risk* and *risks* are 147,099 of its 498,477 matches, and at least 6,632 of the 38,628 matches of *may* are the month.

Data and code availability. The code, the derived data and a runbook that regenerates every number, table and figure are available in the project repository. The source reports are public documents published by the firms.

References

- Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics* 90, 414–427.
- Campbell, D.T., Fiske, D.W., 1959. Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin* 56, 81–105. doi:10.1037/h0046016.
- Hill, M.O., 1973. Diversity and evenness: A unifying notation and its consequences. *Ecology* 54, 427–432. doi:10.2307/1934352.
- Lagasio, V., 2024. ESG-washing detection in corporate sustainability reports. *International Review of Financial Analysis* 96, 103742. doi:10.1016/j.irfa.2024.103742. open access, CC BY 4.0.
- Loughran, T., McDonald, B., 2011. When is a liability not a liability? textual analysis, dictionaries, and 10-ks. *The Journal of Finance* 66, 35–65. doi:10.1111/j.1540-6261.2010.01625.x.
- Manning, C.D., Raghavan, P., Schütze, H., 2008. *Introduction to Information Retrieval*. Cambridge University Press.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* 12, 2825–2830.
- Spärck Jones, K., 1972. A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation* 28, 11–21. doi:10.1108/eb026526.